

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

1st September 2021

Subject SA4 – Pensions and Other Benefits

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1. Please read the instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the statutory laws and accounting standards applicable for the Employee Benefits Schemes. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
- 3. Mark allocations are shown in brackets.*

- Q. 1)** Government of a developing country is providing pension benefits to its employees & X was a department functioning under it. As a part of the administrative reform, the Government decided to establish a State-Owned Corporation (C1) on 1st Jan 1995 and all the existing employees of the department X were transferred to the Corporation. Employees of X department are eligible for a pension equal to 50% final salary payable by death, retirement at age 60 or exit from company after 10 years of service. The pension includes 'Dearness Relief' that is linked to CPI. The maximum pension is limited to INR 75000 per month.

A Pension Trust was formed by the Corporation (C1) to manage and administer this defined benefit (DB) pension scheme. The DB scheme was later closed for new employees on 1st Jan 2005 after a new defined contributions scheme was introduced by the Government. The transferred employees of department X and employees recruited by the corporation between 1995 to 2005 continue to accrue benefits under the DB scheme.

The Government promised to pay to the trust its proportionate share of benefits payable to the transferred employees in respect of their service prior to 1st Jan 1995 on pay-as-you-go basis as and when the employees exit from the services of corporation (C1).

- i)** Discuss the risks faced by the State Government in adopting a pay-as-you-go approach for reimbursement to the corporation as against transferring a one-time lump-sum to the trust to settle the liability. Your answer should include discussion on risks associated with each of the various ways possible to measure the lump-sum amount. (8)

The corporation has been availing services of an Actuary every year for compliance under AS-15. Along with the AS-15 valuation report, the corporation also seeks certification from the Actuary on the contribution rate (as a percentage of pensionable pay) that needs to be paid over the term of obligation. Some of the extracts from the Actuary's reports are as follows (all amounts in INR):

	2020-21	2019-20	2018-19
Number of actives / pensioners	15000 / 19700	17000 / 18500	18000 / 18100
Average Age actives / pensioners	49 / 70	48.8 / 70.1	48.7 / 70.1
Total monthly Pensionable Salary	93 crore	98 crore	90 crore
Total Monthly Pension in payment	84 crore (60 crore Govt portion)	75 crore (52 crore Govt portion)	65 crore (45 crore Govt portion)
Assumptions used Discount rate/ salary growth / pension growth	7.50% / 6.00% / 4.00%	7.10% / 6.00% / 4.00%	6.80% / 6.00% / 4.00%
Term of obligation	12 years	12.3 years	12.5 years
Present Value of Obligation (based on past service)	4465 crore	4565 crore	4260 crore
Present Value of Obligation (based on total service)	5650 crore	6050 crore	5595 crore
Asset Details			
• Investments	2550 crore	2900 crore	3000 crore
• Amounts receivable	750 crore	350 crore	
Contribution Rate (as a % of pensionable salary)	40.00%	38.00%	33.00%
Investment allocation			
• State Govt securities	20%	20%	20%
• Corporate bond	10%	10%	10%
	70%	70%	70%

- | | | | |
|---------------------------|--|--|--|
| • Central Govt securities | | | |
|---------------------------|--|--|--|

- ii) List the valuation methods given in Actuarial Practice Standard 15. Outline the differences between the valuation method prescribed for AS-15 and the method appropriate for determining contribution rate for this scheme. (5)

The contract of the existing Consulting Actuary has expired and you have been appointed as new actuarial advisor by the Trust. Trustees are concerned about the increase in the contribution rate over the years and have approached you to provide your assessment of the financial status of the scheme and reasons for steep increase in contribution rates.

- iii) Discuss in detail various professional and technical aspects to be considered while preparing your response to the trustees. Your answer should also include quantitative assessment for the increase in contributions based on the information provided. (10)

The trustees are keen to reduce the volatility in the contribution rates and to reduce the contribution rates.

- iv) Discuss in detail various measures that can be adopted to meet the above two objectives of the trustees. (7)

It has been further observed that the Government has not been reimbursing its share of benefit obligations as demanded by the trust for the last 2 years and this has been reflected as 'Amounts Receivable from the Government' in the balance sheet of the Trust. The trustees are of the view that the trend of "unpaid Government Obligation" will continue in the coming years also but are confident that the Government will pay all the recoverable amount at some point. Trustees have sought your view on the impact of this situation.

- v) Discuss impact of such circumstances on the scheme. Your answer should cover following aspects:
- Challenges posed to the trust due to these short recoveries.
 - Measures that can be taken to manage the situation.
 - Impact on funding valuation and hence the contribution rate.
 - Impact on accounting valuation under AS-15.
- (10)

There is a proposal to transfer 50% of the active employees of C1 to another State -owned corporation (C2) offering similar pension benefits. The following options are being considered by the Trustees for settling the accrued pension benefits of transferring members:

Option 1: The transferring members will remain in the existing Trust & C2 will pay the annual contribution in respect of these employees and pension will be paid from the current trust when it is due.

Option 2: Transfer an amount corresponding to the accrued benefits to the C2 which will then pay the pension benefits from its own pension trust.

- vi) Discuss the merits & demerits of each option from the perspective of the transferor pension trust (C1)? (10)

[50]

Q. 2) A multi-national Company ABC operating in the travel industry in India sponsors multiple Defined Benefit schemes to its employees. Given below are the benefit details of the schemes:

- **Gratuity:** A lumpsum benefit based with the formula: $21/26 \times \text{last drawn monthly basic salary at exit} \times \text{service}$. The benefit is uncapped and has a vesting criterion of 3 years in case of separation due to voluntary withdrawal and retirement. The scheme is fully funded with a large insurance company in India.
- **Leave encashment:** Benefit is provided on separation where the maximum leave allowed to be accumulated is 120 days. Formula for calculating the leave encashment amount is $\text{last drawn monthly gross salary} \times \text{unavailed leave balance at exit} / 30$.
- **A long-term service benefit** where benefit of INR 10,000 is provided on completion of 5 years, INR 20000 on completion of 10 years, INR 30000 on completion of 15 years, INR 40000 on completion of 20 years and INR 50000 on completion of 25 years.
- **Post-retirement medical benefit** is extended to all retired employees where the retired employees would be provided hospitalization insurance cover of sum insured INR 500,000 after retirement until death. Family of the employees is not covered under this scheme. The benefit is insured.

The auditors are reviewing the actuarial reports prepared for the above schemes on 31st March, 2021 and have requested for a write-up on the actuarial gains and losses.

- i)** State the possible reasons of the actuarial gains / losses arising in each of the above scheme. Also state how the actuarial gains and losses of each of the scheme would be recognized under IND AS19 accounting standards. (8)
- ii)** Given the sudden increase in premium rates, the Company is planning to self-insure the post-retirement medical scheme. State the actuarial & other risks to be considered by the company before taking a decision. (4)
- iii)** The Company, as a part of controlling the expenses, explore several measures; These include reduction in workforce up to 10%, changes in benefit design, closing the non-mandatory schemes to new entrants and temporary reduction in salaries. Accordingly, the Actuary is asked to produce a report to help discuss various strategies:
 - a)** Suggest possible options of design changes to the above four schemes to help the Company strengthen its financial statements. Also discuss on the accounting treatment of these changes under IND AS19 reporting. (12)
 - b)** Instead of changing benefits, the other suggestion by the Management is to reduce the workforce up to 10%. Describe the accounting treatment of the proposed measure to be followed under IND AS19 reporting. (4)
 - c)** Outline any other issues to be considered while taking the above decisions from the perspective of different stake holders. (6)
- iv)** The Company also has an exempt self-managed Provident Fund Trust. The new auditor has now informed the CFO that an actuarial valuation of the same needs to be

conducted. This has come as a surprise since the CFO believes that Provident Fund scheme is a Defined Contribution scheme and has many questions on the need to have actuarial valuation of the scheme. He seeks the opinion of the Actuary on this issue.

- a) Set out the points from Guidance Note 29 / IND AS19 accounting standards and otherwise, explaining to the CFO why the actuarial valuation is required for Provident Fund Scheme. (4)
 - b) Describe the data, assumptions and various valuation methodologies to be used for Provident Fund valuation. (8)
 - c) What are the key risks of self-managing the Provident Fund scheme by the employer? (4)
- [50]**
